SJAA Ephemeris Bulletin

Nov 2025

A Tail of Two Comets

October gave us great views of two relatively bright comets, and the views continue into November. The comets C/2025 R2 Swan and C/2025 A6 Lemmon have put on quite a show, even as R2 Swan's tail has all but disappeared from view, leaving only a faint suggestion of a dust tail. Lemmon, on the other hand, sports a dramatically arcing dust tail, and an ion tail that is changing shape by the minute, producing smoke-like wisps that dance in and out of formation over time.

Come to any of our member observing nights during this month's dark sky window, or join us at our public events to catch glimpses of these two interlopers of the night sky. Look forward to next month's quarterly Ephemeris, which will feature images of these comets from across our very own members.

In addition to football and feasting, November offers up some fine, quality SJAA programming. From Astro 101, to Starry Nights and ITSP, there's lots to be thankful for. I'm thankful for our passionate membership, and all the folks that keep the club running. I bid you all to keep looking up!

Carl Svensson
Vice-President
Ephemeris Editor

Target of the Month: Helix Nebula

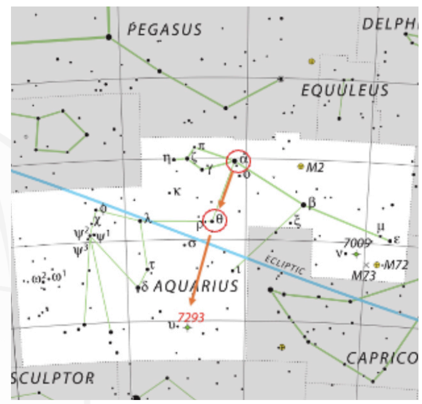
Magnitude	7.3
Size	2.5-4ly
Object Type	Planetary Nebula
Constellation	Aquarius
Distance	660 ly
Age	11,000 years

NGC 7293, better known as the Helix Nebula, is a planetary nebula, named because they resembled planets when viewed through early telescopes. It was discovered by German astronomer Karl Ludwig Harding in September, 1823. The Helix Nebula was formed when a sun-like star reached the end of its life, expanded into a red giant, and then shed its outer layers into space. The remaining hot core is now a white dwarf, which emits ultraviolet radiation that causes the expelled gas to glow, creating the colorful, eye-shaped nebula.

The Helix Nebula is 2.5 to 4 light-years in diameter, making it one of the largest and closest planetary nebulae to Earth. In the sky, it appears about 25 arcminutes in diameter, which is almost half the width of a full moon.

With a magnitude of 7.3, it can be seen with a small telescope or even binoculars under dark skies. A nebula filter will increase the contrast and offer even better views.

To find the Helix, locate the stars Alpha Aquarii (Sadalmelik) and Theta Aquarii (Ancha) in the constellation Aquarius. Draw an imaginary line between the two stars. Extend that line past Ancha for a distance roughly twice the original line's length. The Helix Nebula (NGC 7293) will be in that area.



Top: Helix Nebula location from IAU and Sky & Telescope magazine (Roger Sinnott & Rick Fienberg) Bottom: Helix Nebula image from 10-16-2025 at RCDO by Joe Fragola; collection time of 41 minutes (246 10-second subframes) using a Vaonis Vespera II with dual band filter.

Announcements

Upgraded Email Lists The Observer's email list is an invaluable resource for our members. Members share their observation reports, announcements of upcoming celestial highlights, tips for observing, astro news, and all sorts of interesting astronomy-related tidbits. Along with our Discord server (https://discord.gg/wQ6nZGaJKW), it is a forum for members to connect with one another virtually and to share their passion for astronomy. Soon, we will enable members to manage their subscription to this list through the membership portal (https://membership.sjaa.net) at any time, including when they sign up or renew. This will automate the process of managing the Observer's list, so nobody has to wait around for someone to approve their request. This will also enable us to offer members more lists in the future, so anyone can pick and choose discussion lists based on their interests. Look for more information in your email shortly.

New Renewal System Some of you may have noticed that we have changed how new members sign up, and how returning members renew their memberships. This system provides us with better automation and record keeping, which helps us provide robust and timely services for our members. If you have any questions or face any issues, please reach out to us at asksjaa@sjaa.net. As always, you can reach our renewal and sign up system through the website at https://www.sjaa.net/join-the-sjaa/.

We Always Need More Volunteers There are many volunteer opportunities in SJAA! While we love having seasoned experts in our volunteer ranks, we happily welcome enthusiastic beginners, too. Opportunities range from operating telescopes, to authoring content for publication, to helping with event logistics. Please reach out to us at volunteer@sjaa.net to get started!

SJAA Nov 2025

Sunday	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
						1 Board Mtg 6:00PM Guest Speaker 8:00PM 82%
2 Loaner Exchange / Solar Sunday / FixIt Program 1:30PM DST Ends Harlow Shapley's birthday 90%	3 Tsutomu Seki's birthday 96%	4 99%	5 Fred Whipple's birthday Full Moon 100%	6 Moon 0.8° N of the Pleiades Astro 101 Online 7:00PM 98%	7 Marie Curie's birthday 93%	8 Charles Kowal's birthday 86%
9 Carl Sagan's birthday 77%	10 Jupiter 4° S of Moon Moon 1.7° N of Beehive star cluster 66%	11 Last Quarter Moon Imaging SIG 7:30PM 56%	12 Mercury 1.3° S of Mars Regulus 1° S of Moon 45%	13 James Clerk Maxwell's birthday 34%	14 25%	15 William Herschel's birthday Starry Nights Star Party 6:00PM 17%
16 10%	17 Spica 1.2° N of Moon 5%	18 2%	19 New Moon Foothill College Talk: Enceladus 7:00PM 0%	20 Edwin Hubble's birthday 0%	21 Uranus at opposition Astro 101 6:00PM In-Town Star Party 7:15PM 2%	22 Mendoza Ranch Member Observing 4:30PM 6%
23 11%	24 18%	25 Pluto 0.4° S of Moon Astro 101: Astronomy Beginners' Forum, Online 7:00PM 27%	26 36%	27 First Quarter Moon Anders Celsius' birthday 47%	28 58%	29 Saturn 4° S of Moon Neptune 3° S of Moon Christian Doppler's birthday 69%
30 Kaoru Ikeya's birthday 79%						

Times and dates are subject to change. Please check sjaa.net and www.meetup.com/sj-astronomy for the latest updates.

Meteor Shower

Dark Sky Window